Vision-Proprioception Fusion for Contact Force and Point Estimation

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Abstract

In this project, we study the problem of estimating contact force and contact point during manipulation from multimodal sensory inputs. Vision-only approaches often suffer from severe occlusion and lack direct physical feedback. To address this, we propose a multimodal framework that fuses RGB images and proprioceptive signals to predict contact force and contact point sequences. Specifically, we adapt the Action Chunking with Transformer (ACT) architecture to incorporate joint torque information for contact-aware prediction. We evaluate our method on a custom MuJoCo lid-spinning task with rich contact dynamics. Experimental results demonstrate that our model can effectively recover contact force and point under partial observation, and that larger query sizes lead to more stable and accurate predictions.

1. Introduction

Robotic manipulation tasks that involve contact – from inserting a key into a lock to assembling tight-fitting parts – benefit from combining vision with proprioceptive feedback (e.g. forces/torques and joint sensing). Humans naturally use both vision and touch in such tasks: Visual feedback provides semantic and geometric object properties for accurate reaching or grasp pre-shaping. Haptic feedback provides observations of current contact conditions for accurate localization and control under occlusions" [3]. In other words, vision offers global context (object pose, shape, when contact is about to occur) while proprioceptive/tactile cues indicate how and where contact is happening (forces, slip, pressure). Many recent works frame this multimodal approach as crucial for contact-rich manipulation, since each modality alone is insufficient for robust performance [5]. For example, vision is needed to locate targets and guide motion at a coarse level, but force feedback can fine-tune the interaction once contact occurs (aligning parts, detecting jamming or completion). Overall, fusing vision and proprioception aims to mirror human-level dexterity by exploiting their complementary strengths.

Vision-only approaches lack direct physical feedback (struggling with invisible contacts, force magnitude, slip, etc.), while force-only approaches lack global context and suffer ambiguity in contact localization [2, 5]. This recognition is echoed across recent literature. For instance, a 2024 survey notes that vision-based recognition fails for objects like transparent or uniform items and is affected by lighting, whereas tactile sensing alone cannot perform object localization or handle. The clear consensus is that neither modality alone suffices for robust contact force and point estimation in general settings. By highlighting these limitations, recent papers strengthen the case that fusion of vision and proprioception is needed to achieve reliable contact perception [1].

Recent research strongly advocates for fusing vision and proprioception to improve contact force and contact point estimation in robotic manipulation. By framing vision as providing a broad view of the scene and proprioceptive sensors as providing direct physical measurements, numerous works argue that the two together offer far richer information than either alone [3, 5]. The challenges in achieving this fusion are non-trivial: heterogeneous data streams, temporal alignment, and deciding how to merge cues are active research problems [4]. Accordingly, contemporary solutions are exploring innovative architectures – from coordinated control schemes to end-to-end deep learning models with sensor fusion layers, recurrent units, and transformers that capture cross-modal relationships over time. Early successes in tasks like peg insertion, assembly, and in-hand manipulation demonstrate that multimodal systems can handle delicate contact-rich operations with precision exceeding either modality's solo performance [3].

Despite these promising advances, existing multimodal manipulation approaches are often evaluated in task-specific settings and focus primarily on end-to-end policy learning, rather than explicitly modeling contact forces and contact points as physically meaningful intermediate quantities. As a result, it remains unclear how well such systems generalize to diverse contact conditions or how

accurately they can recover interpretable contact information from raw visual and proprioceptive inputs. Although recent work increasingly emphasizes multimodal fusion, comparatively limited attention has been given to learning explicit contact representations—contact force and contact point—from synchronized sensing streams. Such representations are essential not only for fine-grained control, but also for achieving physically grounded perception and for supporting downstream analysis or model-based reasoning.

Motivated by this gap, we investigate whether a temporally-aware, cross-modal learning architecture can reconstruct contact forces and contact points directly from fused RGB images and proprioceptive states. To this end, we adapt the Action Chunking with Transformer (ACT) framework [9]—originally designed for visuomotor policy learning—to the contact estimation setting, enabling the model to capture temporal structure, multimodal dependencies, and the dynamics of contact-rich interactions. Using a simulated lid-spinning task in MuJoCo, we train and evaluate this adapted ACT model on synchronized visual observations, joint configurations, and joint velocities, with ground-truth contact forces and contact locations serving as supervision.

2. Related Work

2.1. Action chunking with transformer

Zhao et al. [9] propose Action Chunking with Transformer (ACT) to learn temporally structured manipulation policies using a Transformer-based sequence prediction framework. ACT formulates control as an autoregressive sequence prediction problem, where multi-view visual observations are encoded by convolutional backbones and fused with robot proprioception. A set of query tokens and a Transformer encoder–decoder are used to predict a chunk of future actions, enabling the policy to capture long-range temporal dependencies in complex manipulation tasks.

Despite its success in visual imitation learning, ACT is primarily designed for action sequence prediction and does not explicitly model physical interaction quantities such as contact force or contact location. Consequently, it provides limited interpretability of the underlying contact dynamics, especially in scenarios with visual occlusion.

2.2. MuJoCo

MuJoCo [6–8] is a widely used physics engine for model-based control and contact-rich manipulation research. It provides a differentiable rigid-body simulator with smooth contact dynamics, enabling accurate forward dynamics prediction and efficient computation of analytic derivatives for optimization-based control methods such as iLQR, DDP, and MPC. MuJoCo models contacts using a soft-constraint formulation, where signed-distance fields and compliant normal and tangential force models generate continuous forces that support stable simulation and analytic differentiation. This design allows MuJoCo to expose Jacobians of dynamics and cost with respect to states and actions, facilitating gradient-based trajectory optimization in manipulation scenarios involving intermittent contacts.

Although MuJoCo provides detailed physical quantities—including contact impulses, constraint forces, and penetration metrics—the simulator assumes full state observability and perfect geometric knowledge. It therefore does not address the problem of estimating real-world contact forces or contact locations directly from raw sensor observations such as RGB images or proprioception. As a result, despite its accurate physics and differentiability, MuJoCo alone cannot infer latent contact events under visual occlusion or partial observability.

In contrast, our work uses MuJoCo not as a complete information source for control, but as a generator of supervisory signals for multimodal perception. We extract ground-truth world-frame contact force and contact point trajectories from MuJoCo and pair them with RGB and proprioceptive observations. This enables training a sequence model that learns to infer contact dynamics purely from sensor streams, bridging the gap between simulation-provided contact quantities and real robot perception requirements.

3. Method

3.1. ACT-based Contact Estimation Architecture

We adopt the Action Chunking with Transformer (ACT) architecture as the base model and modify it to explicitly predict sequences of contact points and contact forces instead of action commands. The original ACT framework is designed for sequence-to-sequence action prediction from multimodal observations. In our setting, we reinterpret it as a temporal contact estimation model that outputs physically meaningful contact quantities.

Visual Encoding. At each time step, synchronized RGB images from multiple external cameras are first processed by a shared convolutional backbone to extract visual feature embeddings. These visual tokens are augmented with fixed positional embeddings and serve as the visual input to the transformer encoder.

Proprioceptive and Torque Inputs. In addition to visual observations, joint positions, joint velocities, and joint torques are concatenated to form a proprioceptive state vector at each time step. This vector is projected into an embedding space and appended as additional tokens to the transformer encoder input. By explicitly including joint torque, the model is provided with direct physical signals related to contact forces.

Transformer Encoder–Decoder. The fused visual and proprioceptive tokens are processed by a transformer en-

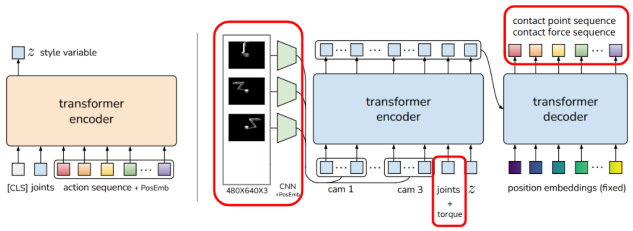


Figure 1. Different from the original ACT, our network uses images collected in Mujoco as visual input and concatenates the joint torque as the proprioceptive input and outputs the sequence of contact force and contact position.

coder to capture cross-modal correlations and long-range temporal dependencies. A transformer decoder then attends to the encoded latent sequence and outputs a sequence of latent contact tokens over a fixed prediction horizon.

Contact Sequence Prediction. Unlike the original ACT model which outputs an action sequence, our modified decoder predicts a sequence of 6D contact states at each time step, consisting of the 3D contact point location and the 3D contact force vector:

$$\hat{\mathbf{y}}_t = [\hat{\mathbf{p}}_t, \hat{\mathbf{f}}_t] \in \mathbb{R}^6.$$

This design enables explicit, temporally consistent reconstruction of contact conditions from raw multimodal sensory inputs.

Through this modification, ACT is repurposed from a policy learning architecture into a contact perception model that jointly models visual context, proprioceptive state, and physical interaction signals.

3.2. Data Collection in MuJoCo

To provide supervision for contact estimation, we construct a multimodal dataset using the MuJoCo physics engine. Each episode records synchronized proprioceptive signals, egocentric RGB images, and ground-truth contact information.

Simulation Environment. We simulate a full hand–object system with accurate actuator dynamics, joint limits, collision geometries, and contact modeling. A combination of scripted trajectories and teleoperation-inspired motions generates diverse interactions such as intermittent touch, sliding, pivoting, and sustained contact. Domain randomization over object pose, texture, lighting, friction, and initial states ensures dataset variability.

Recorded Observations. At each simulation timestep t, we log the proprioceptive state:

$$\mathbf{q}_t, \quad \dot{\mathbf{q}}_t, \quad \boldsymbol{\tau}_t. \tag{1}$$

Additionally, we record images from a *single egocentric RGB camera*, matching the sensing used on the physical robot. Our implementation extracts RGB only; depth images are not captured.

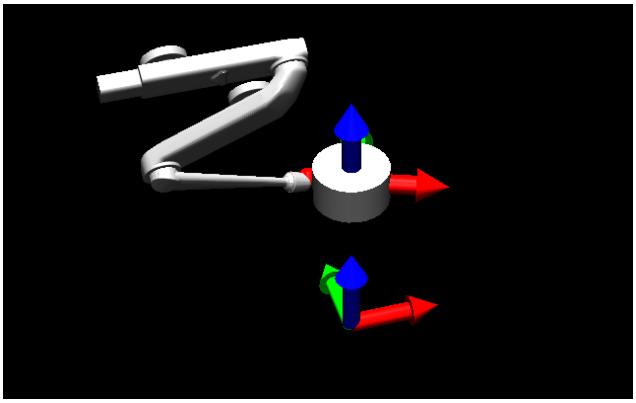


Figure 2. Simulation environment used for multimodal data collection in MuJoCo.

Ground-Truth Contact Extraction. MuJoCo provides detailed contact manifolds including normals, friction forces, impulses, and penetration metrics. From these internal quantities we record the world-frame contact point and force:

$$\mathbf{p}_t^* \in \mathbb{R}^3, \qquad \mathbf{f}_t^* \in \mathbb{R}^3. \tag{2}$$

If no contact exists at timestep t, both vectors are set to zero. These serve as the supervision targets for the ACT-based contact decoder.

Temporal Windowing and Dataset Construction. All modalities are grouped into fixed-length temporal windows for transformer-based sequence modeling:

$$\mathbf{o}_{t:t+H} = \{\mathrm{RGB}_{t:t+H},\ \mathbf{q}_{t:t+H},\ \dot{\mathbf{q}}_{t:t+H},\ \boldsymbol{\tau}_{t:t+H}\}. \tag{3}$$

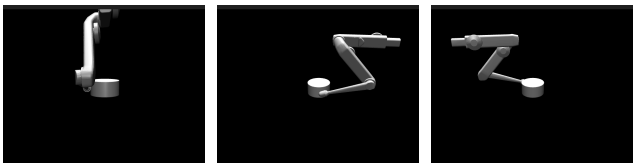
The corresponding contact labels are:

$$\mathbf{y}^*_{t:t+H} = \{[\mathbf{p}_t^*, \mathbf{f}_t^*],\ \ldots,\ [\mathbf{p}_{t+H}^*, \mathbf{f}_{t+H}^*]\}. \tag{4}$$

Data Storage Format. All trajectories are stored using a hierarchical HDF5 (`.h5`) structure. Each episode contains the following datasets:

$$\begin{aligned}
&\mathbf{q}\\
&\dot{\mathbf{q}}\\
&\boldsymbol{\tau}\\
&\mathbf{rgb}\\
&\mathbf{contact_point}\\
&\mathbf{contact_force}
\end{aligned}$$

This format enables efficient indexing, compression, and synchronized retrieval of multimodal sequences.



(a) front camera (b) left camera (c) right camera

Figure 3. Three different cameras for visual input

4. Results

4.1. Effect of Query Size on Contact Estimation

We compare the performance of models trained with different query sizes in terms of both contact point and contact force prediction. As the query size increases, the predicted contact trajectories become progressively smoother and better aligned with the ground-truth signals.

With a single query, the predictions exhibit significant temporal jitter and fail to maintain consistent contact structure over time. Intermediate query sizes (30 and 50) partially recover the overall trend but still suffer from noticeable oscillations and phase lag. Larger query sizes (80 and 100) achieve the most stable and physically consistent predictions, closely matching the ground-truth contact position and force trajectories. The predictions and the ground truth of contact force and contact position for each query are shown in Fig 5

The training and validation loss curves further show that models with larger query sets converge more stably and exhibit reduced variance during training. These results suggest that a sufficiently large query set is essential for capturing the temporal continuity and cross-modal dependencies required for accurate contact reconstruction. And the loss trend during training is shown in Fig 4

5. Discussion

Our experimental findings provide several insights into the role of multimodal fusion and temporal sequence modeling in contact estimation. The most striking result is the strong dependence of reconstruction quality on the query size used in the ACT decoder. As shown in Fig. 4, models trained with a single query (Q = 1) exhibit severe temporal jitter, discontinuous jumps in predicted contact point, and large fluctuations in predicted contact force. These behaviors confirm that contact estimation cannot be treated as a frame-by-frame regression problem: the dynamics of contact are highly coupled over time, and models lacking temporal context fail to maintain physically meaningful structure.

Intermediate query sizes (Q = 30 and Q = 50) partially recover the temporal trends but still suffer from oscillations and delayed alignment with the ground-truth trajectories.

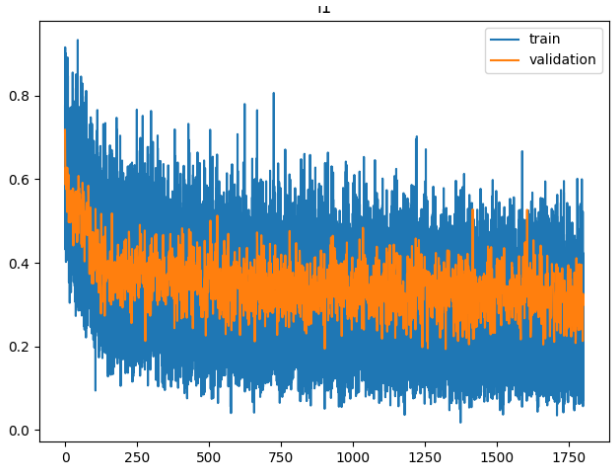


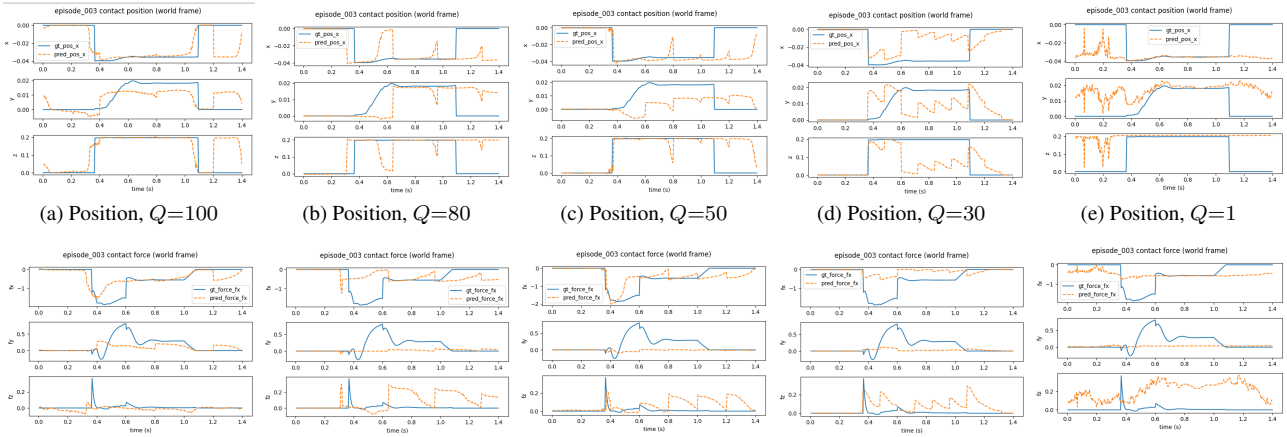
Figure 4. Training and validation loss curves for the ACT model with query size set to 100. The model is trained for 2000 epochs on the MuJoCo contact dataset. Both losses decrease steadily and remain stable throughout training, indicating good convergence and no significant overfitting.

These inconsistencies indicate that short temporal contexts are insufficient for capturing the evolution of contact during lid-spinning, where contact transitions, stick–slip events, and surface interactions unfold over extended time horizons.

In contrast, the larger query models (Q = 80 and Q = 100) yield smooth, stable predictions that closely match the ground-truth contact trajectories. Both contact point and contact force curves exhibit reduced variance and improved temporal coherence, suggesting that larger query sets allow the model to jointly reason about physically plausible sequences rather than isolated snapshots. This result reinforces the view that contact phenomena has long-range temporal dependencies, and that transformer-based architectures benefit substantially from broader receptive fields when modeling such dynamics.

The training and validation loss curves further support this interpretation. As seen in Fig. 3, models with large query sizes converge more stably and exhibit smaller variance between training and validation losses, indicating stronger generalization and reduced sensitivity to noise in the sensory modalities. This suggests that larger query sets help regularize the learning process by encouraging the model to discover consistent temporal patterns rather than overfit to instantaneous fluctuations.

These findings have several implications for multimodal perception in robotics. First, the success of the torque-augmented ACT model highlights the importance of proprioceptive feedback for resolving ambiguities inherent in vision-only systems. Occlusions, featureless textures, and challenging lighting conditions frequently obscure contact



(a) Position, Q=100 (b) Position, Q=80 (c) Position, Q=50 (d) Position, Q=30 (e) Position, Q=1

(f) Force, Q=100 (g) Force, Q=80 (h) Force, Q=50 (i) Force, Q=30 (j) Force, Q=1

Figure 5. Effect of the number of queries Q on contact reconstruction. The top row shows the reconstructed contact **position**, and the bottom row shows the reconstructed contact **force**, under different query settings.

cues, yet proprioceptive signals—especially joint torque—respond immediately and reliably when interaction occurs. Our results show that the transformer learns meaningful cross-modal associations: torque spikes often coincide with changes in predicted contact force, and visual context helps localize where contact is likely occurring.

Second, the improvement observed with higher query sizes suggests that multimodal contact estimation benefits from architectures capable of integrating information across long temporal windows. This is particularly relevant for contact-rich tasks where interactions are intermittent, subtle, or dominated by frictional effects. A model that can encode not just the present sensory state but also the temporal structure leading up to it is better positioned to infer accurate physical quantities.

6. Conclusion

In this work, we presented a multimodal ACT-based framework capable of estimating contact forces and contact points directly from fused RGB and proprioceptive signals. By modifying the ACT decoder to output physically interpretable contact quantities, we demonstrated that temporally aware multimodal transformers can successfully recover contact dynamics under partial observability. Evaluation on a custom MuJoCo lid-spinning dataset shows that larger query sets significantly enhance both temporal consistency and prediction accuracy.

This project highlights the promise of multimodal sequence models for physically grounded perception in robotic manipulation. Future work will expand the dataset to include more diverse dynamic interactions, randomized object geometries, and multi-contact scenarios. Another important direction is cross-embodiment generalization, enabling contact estimation across different robotic hands with minimal fine-tuning. Finally, integrating the contact estimation module into closed-loop manipulation policies may enable more reactive, force-aware control strategies on real hardware.

Overall, our findings provide evidence that vision–proprioception fusion, combined with transformer-based temporal modeling, is a powerful approach for advancing contact-rich manipulation. This work offers a foundation for future research in multimodal perception, sim-to-real transfer, and dexterous robotic interaction.

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